

Name:

Admission No.



Indian Institute of Technology (Indian School of Mines) Dhanbad

Department of Mining Engineering

Makeup Class Test Winter Semester 2023-2024 for 4th Year B.Tech. + 1st Year Ph.D. (14 nos.)

Subject: Deep Coal Mining (Code: MND 412)

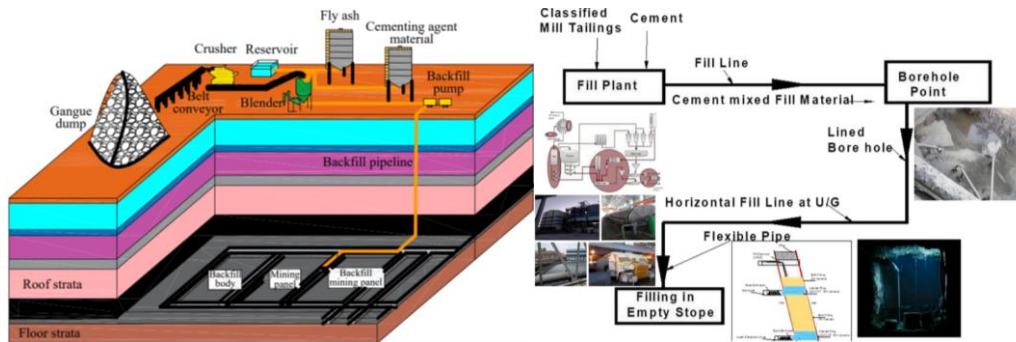
Time: 60 minutes (07:15 PM – 08:15 PM)

Date: 05-05-2024

Maximum Marks: 40

Answer **all** the questions and **marks** are assigned against each question

1. Draw a schematic diagram of backfilling process in an underground coal mine mentioning each operation. (2 marks)



2. Discuss the effects of curing time and binder type and its percentage on stress-strain behavior of paste backfill. (2 marks)

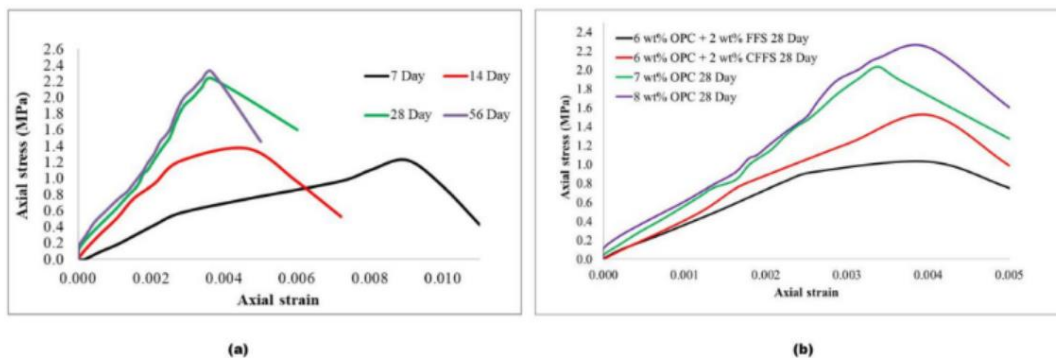


Fig. 11 Typical stress-strain plots: **a** effect of curing time on the stress-strain behaviour of paste backfill (8 wt% OPC) and **b** effect of binder type and dosage on the stress-strain behaviour of paste backfill

3. Briefly explain the uniqueness of Adriyala Longwall Project of Singareni Collieries Company Limited, Telangana. (2 marks)

Punch entry to access coal seam directly from Opencast high wall

11 kV Power transmission to belowground to reduce power interruptions due to voltage drop.

A total of 7.5 km overhead lines and 19 km of 185 sq.mm XLPE cable is laid to meet the huge power requirements.

Auto power factor correction unit was installed for the first time to curtail the wasteful energy loss.

The total connected load of the project was 24 MW and increased to 30 MW with the addition of Air Chilling, Nitrogen and Compressor plants.

Pre-tensioned Cable bolting of 6.1 m long 60 tons capacity for effective roof support and to eliminate standing supports in gate roads

400 kW High capacity fan commissioned to supply 15000 cu.m/min air with high water gauge

Air chilling plant to meet the requirement of Longwall face, outsourced air chilling plant of 2424TR was installed at mouth of Punch Entry-5.

50 cu.m/sec of air quantity, chilled from ambient with 50 to 75% relative humidity to 9°C at discharge point is being supplied, at the entry of PE-5.

From there cooled air is ventilated to Longwall face through a dedicated air way via PE5 and Longwall Main gate.

After commissioning of air chilling plant and supply of cool air, the temperature has been brought down by 3 to 4°C.

Efficiency and effectiveness of men and machinery increased considerably.

Floor Concreting in underground: To facilitate transport of Longwall equipment using diesel vehicles, one of the trunk road ways with weak floor was concreted with M-40 grade self-compacting concrete (SCC) for a length of about 2.5 km with a gradient not more than 1 in 4. Longwall panel top gates were also concreted for about 200 m length near coal floor from trunk roadway to reach of stone floor in bottom section.

Floor Concreting in underground: In gate roadways, stone floor is strong enough to take load of heavy Longwall equipment. It was proposed to do concreting in gates where stone floor disturbed due to water flow to facilitate movement of mine cruiser up to Longwall face. Diesel transport vehicles for speedy & safe transportation of heavy Longwall equipment and other mine material supplies.

High capacity Belt conveyor: Conveyor system of 9.30 km long from Longwall gate belt discharge below ground to Coal Screening Plant (CSP) is installed and commissioned with 11 Conveyor belts with installed capacity of about 12 MW. The conveyor system consists of nine Steel cord belts (800 to 2500 ST) & two PVC belts (Type-15) of 1600 mm width, 3500TPH capacity, 4.0m/s speed and Variable Frequency Drives (VFDs). The system includes one Booster belt or Tripper drive belt conveyor too, in below ground. The above entire system was provided by CODCO.

VFD controlled unmanned belt conveyor system for soft start and power conservation

Automation Systems for sequence control/less manual intervention: The automation of the Longwall face equipment is Shearer Initiated Roof Support Advance (SIRSA) with batch control and auto horizon control.

For automation, CATERPILLAR, using PMC™ system as shown in Fig.2, which offers a tailored control unit for each of the various Longwall functions:

PMC™-R for roof support systems

PMC™-D as drive control (AFC)

PMC™-V for visualization and parameter setup of the drive system

PMC™-P node computer to act as interface between the Cat controller network, third-party systems and the mine computer. The module implements data transmission to the surface via optical fiber, modem or copper wire.

VCU for visualization and control of all Longwall operations at the surface or with an explosionproof computer underground.

Mine Cruiser for faster transport of men

Pan line bolter for speedy roof and rib bolting & use of Plastic wire mesh during salvage of Longwall panel

Bolter Miner for Speedy development of Gate roads for Longwalls

RO plant to supply treated water to improve powered support hydraulics

Chairlift man riding system laid on overburden ramp to facilitate men transport.

High Capacity dewatering pumps (820 HP, 1000 GPM, and 500 m head) introduced for pumping mine water directly to surface.

Membrane type N₂ Plant of 1200 cfm capacity to flush Longwall goaf with inert Nitrogen gas consisting N₂-98% and O₂-2%.

20 point tube bundle system for environment monitoring.

M/s Rod Wheel Technology (Pumping) 400 m³/hr: Slurry water direct delivery to surface form 86AL (by stage wise pumping).

M/s Zamep- Poland, Pumps (pumping) 315 m³/hr water direct delivery to surface form 89L/N4 750m head (by single pump).

4. Define the following terms with a suitable diagram: (a) angle of break (b) angle of draw (c) inflection point. (1 + 1 + 1 = 3 marks)

Inflection point:

The subsidence value at the inflection point is about 1/2 of the maximum subsidence point of the surface. Therefore, the 1/2 surface maximum subsidence point is selected as the inflection point.

The inflection point is often placed over the edge (rib) of an excavated panel and assigned $\frac{s_{max}}{2}$.

Angle of break:

The angle between the vertical and the point of maximum tensile strain from the edge of the underground working.

Angle of draw or Limit Angle:

The angle between the vertical and the edge of the subsidence trough from the edge of the underground working.

5. Three receivers (x_1, x_2, x_3) on a line detect an event (t_1, t_2, t_3) that occurs between receivers 2 and 3. Determine the event location (x_s) and time (t_s) if it is locatable. (5 marks)



Solution: The receivers are located on a line as shown in the sketch.

From the basic distance - time principle

$$(x_s - x_1) = v(t_1 - t_s) \text{ or } x_s + vt_s = x_1 + vt_1$$

$$(x_s - x_2) = v(t_2 - t_s) \text{ or } x_s + vt_s = x_2 + vt_2$$

$$(x_3 - x_s) = v(t_3 - t_s) \text{ or } x_s + vt_s = x_3 - vt_3$$

This system is over determined because there are three equations in only two unknowns. In matrix form the system is

$$\begin{bmatrix} 1 & v \\ 1 & v \\ 1 & -v \end{bmatrix} \begin{Bmatrix} x_s \\ t_s \end{Bmatrix} = \begin{Bmatrix} x_1 + vt_1 \\ x_2 + vt_2 \\ x_3 - vt_3 \end{Bmatrix}$$

Premultiplying by the transpose of the rectangular matrix on the left side gives

$$\begin{bmatrix} 1 & 1 & 1 \\ v & v & v \end{bmatrix} \begin{bmatrix} 1 & v \\ 1 & v \\ 1 & -v \end{bmatrix} \begin{Bmatrix} x_s \\ t_s \end{Bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ v & v & v \end{bmatrix} \begin{Bmatrix} x_1 + vt_1 \\ x_2 + vt_2 \\ x_3 - vt_3 \end{Bmatrix}$$

$$\begin{bmatrix} 3 & v \\ v & 3v^2 \end{bmatrix} \begin{Bmatrix} x_s \\ t_s \end{Bmatrix} = \begin{Bmatrix} (x_1 + vt_1) + (x_2 + vt_2) + (x_3 - vt_3) \\ v(x_1 + vt_1) + v(x_2 + vt_2) - v(x_3 - vt_3) \end{Bmatrix}$$

The inverse of the 2x2 coefficient matrix on the left is $\begin{bmatrix} 3v^2 & -v \\ -v & 3 \end{bmatrix} / 8v^2$

Hence

$$\begin{Bmatrix} x_s \\ t_s \end{Bmatrix} = \frac{1}{8v^2} \begin{bmatrix} 3v^2 & -v \\ -v & 3 \end{bmatrix} \begin{Bmatrix} (x_1 + vt_1) + (x_2 + vt_2) + (x_3 - vt_3) \\ v(x_1 + vt_1) + v(x_2 + vt_2) - v(x_3 - vt_3) \end{Bmatrix}$$

After multiplying the right side, the result is

$$\begin{Bmatrix} x_s \\ t_s \end{Bmatrix} = \begin{Bmatrix} (1/4)(x_1 + x_2) + (1/2)x_3 - (v/4)(t_1 + t_2) - (v/2)t_3 \\ ((1/4v)(x_1 + x_2) + (1/2v)x_3 - (1/4)(t_1 + t_2) - (1/2)t_3) \end{Bmatrix}$$

Comment: In this simple case, the over determined system is readily solved by the procedure of "squaring the system". When many receivers are considered and the event may not be boxed in by the array, the situation is not so easily resolved, although techniques are available for event location in such circumstances.

6. A mine in a flat seam at a depth of 520 m excavates panels with a face width of 260 m and a length of 2,500 m. Mining is full seam height at 4.2 m. The angle of draw and subsidence factor are estimated to be 27° and 0.67, respectively. Determine the (i) maximum subsidence and (ii) trough width for a

single panel, then (iii) sketch the subsidence profile and locate the one-half subsidence points from panel center. **(3 + 2 + 5 = 10 marks)**

The width to depth ratio $W/H = 260/520 = 0.5$, so from the equation associated with Figure 8.15,

$$\frac{S}{h} = a \left\{ 1 - e^{\left[-\frac{1}{2} \left(\frac{W/H}{a/2} \right)^2 \right]} \right\}$$

where the constant a is the subsidence factor, that is, $a = \left(\frac{S}{h} \right)_{max} = S_f$, which is 0.9 in the UK but generally depends on the coal basin stratigraphic sequence.

$$\frac{S}{h} = 0.9 \left\{ 1 - e^{\left[-\frac{1}{2} \left(\frac{0.50}{0.45} \right)^2 \right]} \right\} = 0.415$$

$$S = 0.415 * 4.2 = 1.74 \text{ m}$$

that would be the maximum subsidence over a single panel excavated under UK conditions.

Under the given conditions

$$s' = s \left(\frac{S'_f}{S_f} \right) = 1.74 * \left(\frac{0.67}{0.90} \right) = 1.30 \text{ m}$$

From Table

$\frac{s}{S}$	0.00	0.05	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90	0.95	1.00
$\frac{x}{H}$	0.95	0.59	0.47	0.34	0.28	0.24	0.21	0.17	0.15	0.12	0.08	0.06	0
s	0.00	0.09	0.17	0.35	0.52	0.70	0.87	1.04	1.22	1.40	1.57	1.65	1.74
x	494	306.8	244.4	176.8	145.6	124.8	109.2	88.4	78	62.4	41.6	31.2	0

Thus,

$$\text{For } x = 0.95 * 520, \text{ we have } x' = x \left(\frac{W'_o}{W_o} \right) = (0.95 * 520) \left[\frac{260 + 2 * 520 \tan(27^\circ)}{260 + 2 * 520 \tan(37^\circ)} \right] = 494 \text{ m}$$

$$\text{Similarly for } x = 0.59 * 520 \text{ we have } x' = (0.59 * 520) \left[\frac{260 + 2 * 520 \tan(27^\circ)}{260 + 2 * 520 \tan(37^\circ)} \right] = 306.8 \text{ m}$$

$s_m = \frac{s}{S} * 1.30$	0.00	0.065	0.13	0.26	0.39	0.52	0.65	0.78	0.91	1.04	1.17	1.235	1.3
$x_m = \frac{x}{H} * 520$ $* x \left[\frac{W + 2H \tan(\delta')}{W + 2H \tan(\delta)} \right]$	373.8	232.2	184.9	133.8	110.2	94.4	82.6	66.9	59.0	47.2	31.5	23.6	0

after adjustments for the angle of draw and subsidence factor, Equations (8.37) and (8.38), are made.

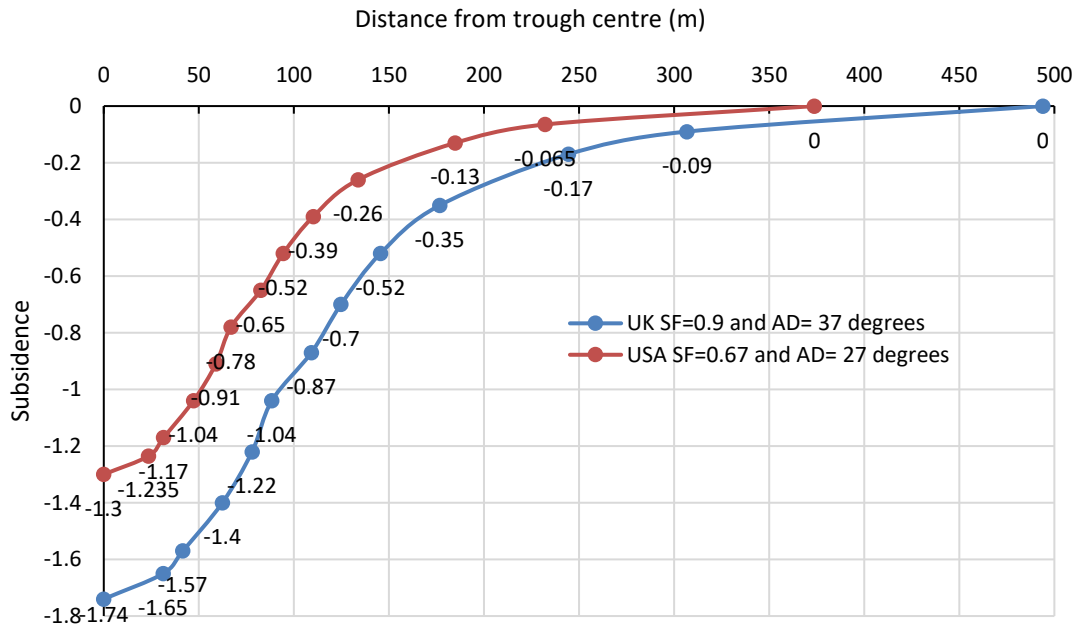
The half-subsidence point occurs where $s/S = 0.5$, that is, where $S = 0.65$ m and $x = 87.3$ m from the panel center.

Trough width is

$$W' = W + 2H \tan(\delta) = 260 + 2 * 520 * \tan(27) = 790 \text{ m}$$

which agrees with the data in the table where zero subsidence occurs at 395 m from the trough center, so the trough width is 790 m.

A plot of the data for the subsidence profile is shown in the sketch.



7. A mine in a flat seam at a depth of 520 m excavates panels with a face width of 260 m and a length of 2,500 m. Mining is full seam height at 4.2 m. The angle of draw and subsidence factor are estimated to be 37° and 0.9, respectively. Determine (i) peak tensile and (ii) compressive strains after completion of a single panel. Also (iii) Determine the secondary compressive strain maximum that may occur at the trough center with strain profile. (iv) Determine the peak tensile and compressive strains after mining three panels side-by-side. (v) Also determine the magnitude of a possible secondary compressive peak strain at the trough center.

(1 + 1 + 4 + 3 + 1 = 10 marks)

The given angle of draw and subsidence factor correspond to UK conditions, so data from Tables can be used directly without adjustment.

The width to depth ratio is 0.5, so,

$$\text{maximum subsidence is } \frac{S}{h} = 0.9 \left\{ 1 - e^{\left[-\frac{1}{2} \left(\frac{0.50}{0.45} \right)^2 \right]} \right\} = 0.415$$

$$S = 0.415 * 4.2 = 1.74 \text{ m}$$

the subsidence depth ratio (S/H) is then $1.74/520 = 3.346 \times 10^{-3}$

According to Table, $E+ = 0.80$ and $E- = 1.35$.

Hence the peak tensile and compressive strains are

$$(i) E_t = (E+) \left(\frac{S}{H} \right) = 0.80 * 3.346 * 10^{-3} = 2.677 * 10^{-3}$$

$$(ii) E_c = (E-) \left(\frac{S}{H} \right) = 1.35 * 3.346 * 10^{-3} = 4.517 * 10^{-3}$$

(iii) there is indeed a secondary compressive strain peak e_c at the trough center.

The magnitude of this peak may be obtained from $\frac{-e}{-E} = e^{\left[-4 \left(\frac{W}{H} - 0.5 \right)^2 \right]} = e^{\left[-4(0.5-0.5)^2 \right]} = 1$

$\therefore e_c = 4.517 * 10^{-3}$ which is compressive

(iv) In this configuration, the total panel width to depth ratio is $(3)(260)/520 = 1.5$ and maximum subsidence, $(0.9)(4.2) = 3.78 \text{ m}$.

The subsidence to depth ratio is $3.78/520 = 7.269 \times 10^{-3}$

From Table 8.3, $E+$ and $E-$ are 0.65 and 0.51, respectively.

Hence the peak tensile and compressive strains are

$$E_t = (E+) \left(\frac{S}{H} \right) = 0.65 * 7.269 * 10^{-3} = 4.725 * 10^{-3}$$

$$E_c = (E-) \left(\frac{S}{H} \right) = 0.51 * 7.269 * 10^{-3} = 3.707 * 10^{-3}$$

(v) The panel width is supercritical so no secondary peak of compression is expected at the trough center about 0.02 times E_c .

8. A circular vertical shaft is planned to have a finished, inside diameter of 5.8 m in an underground hard rock mine. Rock properties are:

$$C_o = 163.5 \text{ MPa}, T_o = 10.21 \text{ MPa}, E = 36.48 \text{ GPa}, \nu = 0.27, \gamma = 25.63 \text{ kN/m}^3.$$

The premining stress state relative to compass coordinates is:

$$S_E = 2414 + 4.53h, S_N = 2897 + 7.92h, S_V = 25.34h$$

where E, N, V refer to compass coordinates (x = east, y = north, z = up), h is depth in m, stress is in kPa and compression is positive. Premining shear stresses are nil relative to compass coordinates. Properties of concrete are: Compressive strength (C) = 39.6 MPa, Tensile strength (T) = 2.93 MPa, Young's modulus (E) = 32.76 GPa, Poisson's ratio (ν) = 0.25. Find:

(a) the best shaft orientation

(b) the unlined shaft wall safety factors in tension and compression at a depth of 1,295 m,

(c) concrete liner thickness in the vicinity of a water-bearing formation where the pressure in the undisturbed ground is 0.621 MPa. Note: The maximum allowable compression in the concrete is 24.14 MPa.

(2 + 2 + 2 = 6 marks)

$$S_E = 2414 + 4.53h = 2414 + 4.53 * 1295 = 8280.35 \text{ kPa} = 8.28 \text{ MPa} = \sigma_E$$

$$S_N = 2897 + 7.92h = 2897 + 7.92 * 1295 = 13153.4 \text{ kPa} = 13.15 \text{ MPa} = \sigma_N$$

$$S_V = 25.34h = 25.34 * 1295 = 32815. \text{ kPa} = 32.81 \text{ MPa}$$

$$M = \frac{8.28}{13.15} = 0.63$$

$$K_{min} = 1 + M - 2(1 - M) = 1 + 0.63 - 2(1 - 0.63) = 0.89$$

$$K_{max} = 1 + M - 2(1 - M) = 1 + 0.63 + 2(1 - 0.63) = 2.37$$

$$FS_c = \frac{C_o}{\sigma_c} = \frac{C_o}{K_c \sigma_1} = \frac{163.5}{2.37 * 13.15} = 5.25$$

$$FS_t = \frac{C_o}{\sigma_t} = \frac{C_o}{K_t \sigma_1}; K_t \text{ is nil, } \therefore \text{ tension n.a. } FS_t \text{ n.a.}$$

$$FS_c = \frac{C_o}{\sigma_v} = \frac{163.5}{32.81} = 4.99$$

Maximum allow C_o in concrete= 3500

$$\text{But also, } \sigma_{max} = \frac{2p}{1 - \left(\frac{a}{b}\right)^2}$$

$$24.14 = \frac{2 * 0.621}{1 - \left(\frac{2.9}{b}\right)^2}; a = \frac{ID}{2}$$

$$\therefore 1 - \left(\frac{2.9}{b}\right)^2 = 0.05145$$

$$\therefore \left(\frac{2.9}{b}\right)^2 = 0.94855$$

$$\therefore \frac{2.9}{b} = 0.9739$$

$$\therefore b = 2.977$$

$$\therefore h = b - a = 2.977 - 2.900 = 0.078 \text{ mm} = 7.8 \text{ cm}$$

Important Formulae: where symbols have its own meaning

$$p_b = C_o \left[\frac{1 - \left(\frac{a}{b}\right)^2}{2FS_c} \right] \quad h = b \left[1 - \sqrt{\left(\frac{C_o - 2p_b FS}{C_o}\right)} \right]$$

$$\frac{S'}{S} = \left\{ 1 - e^{[-2.70\left(\frac{x}{H}\right)]^2} \right\}$$

$$\frac{S}{h} = 0.9 \left\{ 1 - e^{[-2.47\left(\frac{x}{H}\right)]^2} \right\}$$

$$\frac{-e}{-E} = e^{[-4\left(\frac{W}{H} - 0.5\right)^2]}$$

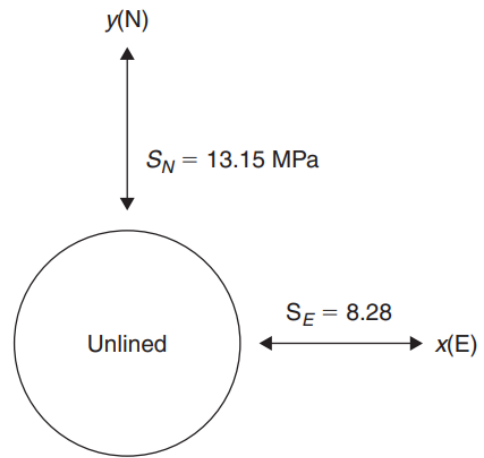


Table 1: Subsidence profile points at various width to depth ratios (from SHB, 1975)

s/S W/H	0.00	0.05	0.10	0.20	0.30	0.40	0.50	X/H	0.60	0.70	0.80	0.90	0.95	1.00
0.20	0.80	0.64	0.57	0.48	0.41	0.37	0.32		0.28	0.23	0.19	0.13	0.08	0.00
0.24	0.82	0.62	0.53	0.43	0.36	0.32	0.28		0.24	0.20	0.16	0.11	0.07	0.00
0.28	0.84	0.61	0.51	0.39	0.33	0.28	0.24		0.21	0.18	0.14	0.09	0.07	0.00
0.30	0.85	0.61	0.50	0.38	0.32	0.27	0.23		0.20	0.17	0.13	0.09	0.06	0.00
0.34	0.87	0.60	0.49	0.36	0.30	0.25	0.22		0.19	0.16	0.12	0.08	0.06	0.00
0.38	0.89	0.60	0.48	0.35	0.29	0.24	0.21		0.18	0.15	0.12	0.08	0.06	0.00
0.40	0.90	0.59	0.47	0.34	0.28	0.24	0.21		0.18	0.15	0.12	0.08	0.06	0.00
0.44	0.92	0.59	0.47	0.33	0.28	0.23	0.20		0.17	0.15	0.12	0.08	0.06	0.00
0.48	0.94	0.59	0.47	0.33	0.28	0.23	0.20		0.17	0.15	0.12	0.08	0.06	0.00
0.50	0.95	0.59	0.47	0.34	0.28	0.24	0.21		0.17	0.15	0.12	0.08	0.06	0.00
0.54	0.97	0.59	0.47	0.34	0.29	0.25	0.21		0.18	0.15	0.12	0.08	0.06	0.00
0.58	0.99	0.59	0.47	0.35	0.30	0.25	0.22		0.18	0.16	0.13	0.09	0.06	0.00
0.60	1.00	0.59	0.47	0.36	0.30	0.26	0.22		0.19	0.16	0.13	0.09	0.06	0.00
0.64	1.02	0.59	0.48	0.37	0.31	0.27	0.23		0.20	0.17	0.13	0.09	0.06	0.00
0.68	1.04	0.60	0.49	0.38	0.32	0.28	0.24		0.21	0.17	0.14	0.10	0.07	0.00
0.70	1.05	0.60	0.49	0.39	0.33	0.29	0.25		0.21	0.18	0.14	0.10	0.07	0.00
0.74	1.07	0.61	0.50	0.40	0.34	0.30	0.26		0.23	0.19	0.15	0.10	0.07	0.00
0.78	1.09	0.63	0.52	0.42	0.36	0.32	0.28		0.24	0.20	0.16	0.11	0.08	0.00
0.80	1.10	0.63	0.52	0.42	0.36	0.32	0.28		0.25	0.21	0.17	0.11	0.08	0.00
0.84	1.12	0.65	0.54	0.44	0.38	0.34	0.30		0.26	0.22	0.18	0.12	0.09	0.00
0.88	1.14	0.67	0.56	0.45	0.40	0.36	0.32		0.28	0.24	0.20	0.13	0.10	0.00
0.90	1.15	0.68	0.57	0.46	0.40	0.36	0.32		0.29	0.25	0.20	0.14	0.10	0.00
0.94	1.17	0.69	0.58	0.48	0.42	0.38	0.34		0.31	0.26	0.22	0.16	0.11	0.00
0.98	1.19	0.71	0.60	0.50	0.44	0.40	0.36		0.33	0.28	0.24	0.17	0.12	0.00
1.00	1.20	0.72	0.61	0.51	0.45	0.41	0.37		0.33	0.29	0.24	0.18	0.13	0.00
1.10	1.25	0.77	0.65	0.55	0.50	0.45	0.42		0.38	0.34	0.29	0.21	0.16	0.00
1.20	1.30	0.81	0.70	0.60	0.54	0.50	0.46		0.42	0.38	0.33	0.25	0.19	0.00
1.30	1.35	0.86	0.75	0.65	0.59	0.55	0.51		0.47	0.43	0.38	0.30	0.23	0.00
1.40	1.40	0.91	0.80	0.70	0.64	0.60	0.56		0.52	0.48	0.43	0.35	0.27	0.00
1.60	1.50	1.01	0.90	0.80	0.74	0.70	0.66		0.62	0.58	0.53	0.45	0.37	0.05
1.80	1.60	1.11	1.00	0.90	0.84	0.80	0.76		0.72	0.68	0.63	0.55	0.47	0.10
2.00	1.70	1.21	1.09	0.99	0.94	0.90	0.86		0.82	0.78	0.73	0.65	0.57	0.16
2.20	1.80	1.31	1.19	1.09	1.04	1.00	0.96		0.92	0.88	0.83	0.75	0.67	0.23
2.40	1.90	1.41	1.29	1.19	1.14	1.10	1.06		1.02	0.98	0.93	0.85	0.77	0.31
2.60	2.00	1.51	1.39	1.29	1.24	1.19	1.16		1.12	1.08	1.03	0.95	0.87	0.41

Table 2: Peak strain multipliers.

W/H	E+	E-
0.10	0.25	2.40
0.20	0.52	2.25
0.30	0.70	2.00
0.40	0.77	1.73
0.50	0.80	1.35
0.60	0.75	1.06
0.70	0.68	0.86
0.80	0.65	0.70
0.90	0.65	0.60
1.00	0.65	0.55
1.10	0.65	0.54
1.20	0.65	0.53
1.30	0.65	0.52
1.40	0.65	0.51

Note

a Tension (+),compression (-).

Table 3: Strain profile points at various width to depth ratios (from SHB, 1975)

e/E W/H	0.00	0.20	0.40	0.60	0.80	1.00	0.80	0.60	0.40	0.20	0.00	0.20	0.40	0.60	0.80	1.00	0.80	0.60	0.40	0.20	0.00
										X/H											
0.20	0.80	0.74	0.69	0.66	0.61	0.49	0.42	0.32	0.27	0.23	0.18	0.12	0.00	0.00	0	0	0	0	0	0	0
0.24	0.82	0.70	0.63	0.60	0.54	0.44	0.37	0.28	0.23	0.20	0.15	0.10	0.00	0.00	0	0	0	0	0	0	0
0.28	0.84	0.66	0.58	0.54	0.49	0.39	0.33	0.24	0.21	0.17	0.13	0.08	0.00	0.00	0	0	0	0	0	0	0
0.32	0.86	0.63	0.55	0.49	0.45	0.35	0.30	0.22	0.19	0.16	0.12	0.07	0.00	0.00	0	0	0	0	0	0	0
0.36	0.88	0.62	0.53	0.46	0.42	0.33	0.29	0.21	0.18	0.15	0.11	0.07	0.00	0.00	0	0	0	0	0	0	0
0.40	0.90	0.61	0.52	0.45	0.40	0.32	0.28	0.21	0.18	0.15	0.11	0.07	0.00	0.00	0	0	0	0	0	0	0
0.44	0.92	0.60	0.51	0.43	0.39	0.31	0.27	0.20	0.18	0.15	0.11	0.07	0.01	0.00	0	0	0	0	0	0	0
0.48	0.94	0.60	0.51	0.43	0.38	0.31	0.27	0.20	0.18	0.15	0.12	0.08	0.01	0.00	0	0	0	0	0	0	0
0.52	0.96	0.60	0.51	0.43	0.38	0.32	0.27	0.21	0.18	0.16	0.12	0.09	0.02	0.00	0	0	0	0	0	0	0
0.56	0.98	0.61	0.51	0.44	0.39	0.33	0.28	0.22	0.19	0.17	0.13	0.10	0.03	0.00	0	0	0	0	0	0	0
0.60	1.00	0.62	0.52	0.45	0.40	0.34	0.29	0.22	0.20	0.18	0.14	0.11	0.05	0.00	0	0	0	0	0	0	0
0.64	1.02	0.63	0.53	0.46	0.41	0.35	0.31	0.23	0.21	0.19	0.15	0.12	0.06	0.00	0	0	0	0	0	0	0
0.68	1.04	0.64	0.54	0.47	0.43	0.37	0.32	0.24	0.22	0.20	0.16	0.13	0.07	0.00	0	0	0	0	0	0	0
0.72	1.06	0.66	0.55	0.49	0.44	0.38	0.34	0.26	0.24	0.21	0.17	0.15	0.09	0.00	0	0	0	0	0	0	0
0.76	1.08	0.67	0.57	0.51	0.46	0.40	0.36	0.27	0.25	0.22	0.19	0.16	0.10	0.01	0	0	0	0	0	0	0
0.80	1.10	0.69	0.58	0.53	0.48	0.42	0.37	0.29	0.26	0.24	0.20	0.17	0.11	0.02	0	0	0	0	0	0	0
0.84	1.12	0.71	0.60	0.54	0.49	0.44	0.39	0.30	0.28	0.25	0.22	0.19	0.13	0.04	0	0	0	0	0	0	0
0.88	1.14	0.73	0.62	0.56	0.51	0.46	0.41	0.32	0.29	0.27	0.24	0.21	0.15	0.05	0.01	0	0	0	0	0	0
0.92	1.16	0.75	0.64	0.58	0.53	0.47	0.43	0.34	0.31	0.29	0.25	0.22	0.16	0.07	0.02	0	0	0	0	0	0
0.96	1.18	0.77	0.66	0.60	0.55	0.49	0.45	0.35	0.33	0.30	0.27	0.24	0.18	0.09	0.04	0	0	0	0	0	0
1.00	1.20	0.79	0.68	0.62	0.57	0.51	0.47	0.37	0.35	0.32	0.29	0.26	0.20	0.10	0.05	0	0	0	0	0	0
1.20	1.30	0.88	0.77	0.71	0.66	0.61	0.56	0.46	0.44	0.41	0.38	0.35	0.29	0.20	0.13	0.07	0.02	0	0	0	0
1.40	1.40	0.98	0.87	0.81	0.75	0.70	0.66	0.56	0.53	0.51	0.48	0.45	0.39	0.30	0.23	0.17	0.1	0	0	0	0
1.60	1.50	1.08	0.97	0.91	0.85	0.80	0.76	0.66	0.63	0.61	0.58	0.55	0.49	0.40	0.33	0.27	0.2	0.03	0	0	0
1.80	1.60	1.17	1.07	1.01	0.95	0.90	0.86	0.76	0.73	0.71	0.68	0.65	0.59	0.50	0.43	0.37	0.3	0.1	0	0	0
2.00	1.70	1.28	1.17	1.11	1.05	1.00	0.96	0.86	0.84	0.81	0.78	0.75	0.69	0.60	0.53	0.47	0.4	0.2	0	0	0
2.20	1.80	1.38	1.27	1.21	1.15	1.10	1.06	0.96	0.94	0.91	0.88	0.85	0.79	0.70	0.63	0.57	0.5	0.3	0	0	0
2.40	1.90	1.48	1.37	1.31	1.25	1.20	1.16	1.06	1.04	1.01	0.98	0.95	0.89	0.80	0.73	0.67	0.6	0.4	0	0	0
2.60	2.00	1.58	1.47	1.41	1.36	1.30	1.26	1.16	1.14	1.11	1.08	1.05	0.99	0.90	0.83	0.77	0.7	0.5	0	0	0